MIL-T-45886A(AR)
25 May 1988
SUPERSEDING
MIL-T-45886
16 August 1976

MILITARY SPECIFICATION

TIMER, SEQUENTIAL: M42E2

This specification is approved for use by U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one Timer, Sequential: M42E2, which is used in the Lance warhead section (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

*2.1.1 Specifications and standards. Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1135

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-C-3098	- Crystal, unit, Quartz, General Specification for.
MIL-I-6866	- Inspection, Penetrant Method of.
MIL-M-38510	- Microcircuit, General Specification for.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for.
MIL-P-50884	- Printed wiring, flexible and rigid-flex.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-275	- Printed Wiring for Electronic Equipment.
MIL-STD-2000 (1A, 2, 3, and 4)	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.
MIL-STD-2118	- Flexible and rigid-flex printed wiring for electronic equipment, design requirements for.

*2.1.2 Other Government documents, drawings and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

12551040	- Timer, Sequential: M42E2
12551050	- Circuit Board, Microprocessor
12551053	- Circuit Board, Output Driver
12551061	- Circuit Board, Switch Routing
12551049	- Component Assembly Board, Micro
12551052	- Component Assembly Board, Output
12551060	- Component Assembly Board, Switch
12551042	- Housing
12551080	- Front Plate
12551081	- Bezel
12551082	- Connector, Electrical Type LJT17MC
12551045	- Switch, Optical

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DOD adapted shall be the issue listed in the current DOD ISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D 3951-82 - Practice for Commercial Packaging.

(Applications for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA., 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and finishes. The contractor shall furnish objective evidence (see 6.6) that the material complies with the specifications listed in the applicable drawing.

*3.2 Assembly. The sequential timer shall comply with all requirements specified on the drawing (dwg) 12551040, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.3).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, AMCCOM, ATTN: AMSMC-QAN-Q(D), Picatinny Arsenal, New Jersey, 07806-5000. The letter of request shall include:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

a. Exact nomenclature of the item as specified on the applicable drawing.

b. Name of the contractor.

c. Contract number or procurement request order number.

d. Number of serial codes required (unused code numbers shall be returned).

e. Security classification if the quantity of six letter codes requested reveals classified production quantities.

3.5 Coded serialization requirements (see 6.2.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

*3.6 Continuity loop. The connector shall show a maximum resistance of 20 milliohms cross pins 24 and 39, and 50 milliohms across pins 40, 41, 4 and 25. Testing shall be as specified in 4.5.1.

3.7 Input power.

*3.7.1 Input current. Channels A and B of the integrated circuit portion of the timer circuit shall draw 400 milliamperes (ma) maximum and channel C shall draw 235 ma maximum when an input voltage of 6.5 ± 0.25 volts direct current (vdc) is applied. Testing shall be as specified in 4.5.2.

*3.7.2 Input voltage. The timer shall operate from a 6.5 + 0.75 to 6.5 - 0.50 vdc interrupted power source. Channels A, B and C shall be powered by separate input power lines that can be interrupted during test.

*3.8 Output current. Each output current pulse and duration at the extreme temperatures of 140 ± 5°F shall be limited to the magnitudes as specified in Table I when tested as specified in 4.5.3.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

TABLE I. - Output current

Signal Name	Connector Pin	- to -	Return Pin	Current (Amperes)
T1A	7		16 or 9	4 ± 0.4
T1B	6		16 or 9	4 ± 0.4
T2A	18		30 or 31	2 ± 0.2
T2B	19		30 or 31	2 ± 0.2
T3A	51		30 or 31	2 ± 0.2
T3B	45		30 or 31	2 ± 0.2
TDCOA	20		16 or 9	2 ± 0.2
TDCOB	21		16 or 9	2 ± 0.2
TSCO	50		16 or 9	2 ± 0.2
TDEA	12		15	0.100 ± 0.02
TDEB	11		14	0.100 ± 0.02

*3.9 Signal output. The sequential timer, as described in 11251040, is a dual channel output device (with the exception of the single channel TSCO signal) wherein, all outputs are drived from three input sections of information. As long as two out of the three input sections provide coincident data, there will be an output. Time pulses shall be as specified in Table II. Testing shall be as specified in 4.5.4.

TABLE II. - Signal output

Test Cycle	Switch Identi	Pos	Time Identi	Pulse Sec	Accuracy Sec	Test Code No. (see 3.5) Chan A		Chan B	
A	T1 h	S*	TDE	1	±0.1	060	001	060	101
	T1 t	O	T1	S	N/A	060	002	060	102
	T1 u	O	T2	S	N/A	060	003	060	103
	TDCO/SCO	R	T3	S	N/A	060	004	060	104
			TDCO	S	N/A	060	005	060	105
			TSCO	S	N/A	060	006		
*S denotes Safe Position									
B	T1 h	O	TDE	1	±0.1	060	007	060	107
	T1 t	S	T1	S	N/A	060	008	060	108
	T1 u	O	T2	S	N/A	060	009	060	109
	TDCO/SCO	R	T3	S	N/A	060	010	060	110
			TDCO	S	N/A	060	011	060	111
			TSCO	S	N/A	060	012		
C	T1 h	O	TDE	1	±0.1	060	013	060	113
	T1 t	O	T1	S	N/A	060	014	060	114
	T1 u	S	T2	S	N/A	060	015	060	115
	TDCO/SCO	R	T3	S	N/A	060	016	060	116
			TDCO	S	N/A	060	017	060	117
			TSCO	S	N/A	060	018	060	118

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

TABLE II. - Signal output (con't)

Test Cycle	Switch Identi	Pos	Time Identi	Pulse Sec	Accuracy Sec	Test Code No. (see 3.5) Chan A		Chan B	
D	T1 h	O	TDE	1	+0.1	060	019	060	119
	T1 t	O	T1	S	N/A	060	020	060	120
	T1 u	O	T2	S	N/A	060	021	060	121
	TDCO/SCO	S	T3	S	N/A	060	022	060	122
			TDCO	S	N/A	060	023	060	123
			TSCO	S	N/A	060	024		
E	T1 h	O	TDE	1	+0.1	060	025	060	125
	T1 t	1	T1	S	N/A	060	026	060	126
	T1 u	5	T2	S	N/A	060	027	060	127
	TDCO/SCO	R	T3	S	N/A	060	028	060	128
			TDCO	S	N/A	060	029	060	129
			TSCO	S	N/A	060	030		
F	T1 h	O	TDE	1	+0.1	060	031	060	131
	T1 t	1	T1	16	±0.0080	060	032	060	132
	T1 u	6	T2	19	±0.1	060	033	060	133
	TDCO/SCO	P	T3	22	±0.1	060	034	060	134
			TDCO	45	±0.1	060	035	060	135
			TSCO	55	±0.0275	060	036		
G	T1 h	O	TDE	1	+0.1	060	037	060	137
	T1 t	2	T1	22	±0.0110	060	038	060	138
	T1 u	2	T2	25	±0.1	060	039	060	139
	TDCO/SCO	R	T3	28	±0.1	060	040	060	140
			TDCO	56	±0.1	060	041	060	141
			TSCO	66	±0.0330	060	042		
H	T1 h	O	TDE	1	+0.1	060	043	060	143
	T1 t	3	T1	33	±0.0165	060	044	060	144
	T1 u	3	T2	36	±0.1	060	045	060	145
	TDCO/SCO	T	T3	39	±0.1	060	046	060	146
			TDCO	67	±0.1	060	047	060	147
			TSCO	77	±0.0385	060	048		
I	T1 h	O	TDE	1	+0.1	060	049	060	149
	T1 t	4	T1	44	±0.022	060	050	060	150
	T1 u	4	T2	47	±0.1	060	051	060	151
	TDCO/SCO	U	T3	50	±0.1	060	052	060	152
			TDCO	78	±0.1	060	053	060	153
			TSCO	88	±0.0440	060	054		
J	T1 h	O	TDE	1	+0.1	060	055	060	155
	T1 t	5	T1	55	±0.0275	060	056	060	156
	T1 u	5	T2	58	±0.1	060	057	060	157
	TDCO/SCO	V	T3	61	±0.1	060	058	060	158
			TDCO	89	±0.1	060	059	060	159
			TSCO	99	±0.0495	060	060		

Source: https://assist.dla.mil -- Downloaded: 202640-22T04:05Z
Check the source to verify that this is the current version before use.

TABLE II. - Signal output (con't)

Test Cycle	Switch Identi	Pos	Time Identi	Pulse Sec	Accuracy Sec	Test Code No. (see 3.5) Chan A		Chan B	
K	T1 h	O	TDE	1	+0.1	060	061	060	161
	T1 t	6	T1	66	±0.0330	060	062	060	162
	T1 u	6	T2	69	±0.1	060	063	060	163
	TDCO/SCO	W	T3	72	±0.1	060	064	060	164
			TDCO	100	±0.1	060	065	060	165
			TSCO	110	±0.0550	060	066		
L	T1 h	O	TDE	1	±0.1	060	067	060	167
	T1 t	7	T1	77	±0.0385	060	068	060	168
	T1 u	7	T2	80	±0.1	060	069	060	169
	TDCO/SCO	X	T3	83	±0.1	060	070	060	170
			TDCO	110	±0.1	060	071	060	171
			TSCO	120	±0.0600	060	072		
M	T1 h	O	TDE	1	±0.1	060	073	060	173
	T1 t	8	T1	88	±0.0440	060	074	060	174
	T1 u	8	T2	91	±0.1	060	075	060	175
	TDCO	N	T3	94	±0.1	060	076	060	176
			TDCO	N/A		060	077	060	177
			TSCO	N/A		060	078		
N	T1 h	O	TDE	1	±0.1	060	079	060	179
	T1 t	9	T1	99	±0.0495	060	080	060	180
	T1 u	9	T2	102	±0.1	060	081	060	181
	TDCO/SCO	U	T3	105	±0.1	060	082	060	182
			TDCO	78	±0.1	060	083	060	183
			TSCO	88	±0.0650	060	084		
O	T1 h	1	TDE	15	±0.1	060	085	060	185
	T1 t	O	T1	100	±0.0495	060	086	060	186
	T1 u	O	T2	103	±0.1	060	087	060	187
	TDCO/SCO	Y	T3	106	±0.1	060	088	060	188
			TDCO	120	±0.1	060	089	060	189
			TSCO	130	±0.0650	060	090		
P	T1 h	2	TDE	15	±0.1	060	091	060	191
	T1 t	2	T1	221	±0.1105	060	092	060	192
	T1 u	1	T2	224	±0.1	060	093	060	193
	TDCO/SCO	Z	T3	227	±0.1	060	094	060	194
			TDCO	130	±0.01	060	095	060	195
			TSCO	140	±0.0700	060	096		

Disconnect Power from J1 Connector Pin 26

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

TABLE II. - Signal output (con't)

Test Cycle	Switch Identi	Pos	Time Identi	Pulse Sec	Accuracy Sec	Test Code No. (see 3.5) Chan	A	Chan	B
Q	T1 h	O	TDE	1	±0.1	060	200	N/A	
	T1 t	2	T1	22	±0.0110	060	201	N/A	
	T1 u	2	T2	25	±0.1	060	202	N/A	
	TDCO/SCO	R	T2	28	±0.1	060	203	N/A	
			TDCO	56	±0.1	060	204	N/A	
			TSCO	66	±0.0330	060	205	N/A	
Reconnect Power to J1 Connector Pin 26 Disconnect Power from J1 Connector Pin 27									
R	T1 h	O	TDE	1	±0.1	N/A		060	206
	T1 t	2	T1	22	±0.0110	N/A		060	207
	T1 u	2	T2	25	±0.1	N/A		060	208
	TDCO/SCO	R	T3	28	±0.1	N/A		060	209
			TDCO	56	±0.1	N/A		060	210
			TSCO	66	±0.0330	N/A		060	211
Reconnect Power to J1 Connector Pin 27 Disconnect Power from J1 Connector, Pins 27 and 33									
S	T1 h	O	TDE	S	N/A	060	212	060	218
	T1 t	2	T1	S	N/A	060	213	060	219
	T1 s	2	T2	S	N/A	060	214	060	220
	TDCO/SCO	R	T3	S	N/A	060	214	060	221
			TDCO	S	N/A	060	216	060	222
			TSCO	S	N/A	060	217	060	223
Reconnect Power to J1 Connector, Pins 27 and 33									

Abbreviation Key for Table II:

Cyc	Cycle
Identi	Identification
Pos	Position
Sec	Seconds
Test No.	Test Identification Number
Chan	Channel
h	Hundreds
t	Tens
u	Units
N/A	Not Applicable
s	Safe, No Signal Event

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

*3.10 Environmental testing (see 4.5.5). The unit shall environmentally meet all the requirements of the Lance Stockpile to Target Sequence (STS), some of which are classified. Those unclassified environments required for First Article Testing and are applicable to in process screening procedures and are reiterated below.

3.10.1 Low temperature. The sequential timer shall withstand exposure to a chamber temperature of -40 ± 5 degrees Fahrenheit (°F) for a minimum of four (4) hours without evidence of external damage (see 6.4) and shall be tested as specified in 4.5.5.1 for compliance with 3.7.1 and 3.9.

*3.10.2 High temperature. The sequential timer shall withstand exposure to a chamber temperature of 140 ± 5 degrees Fahrenheit (°F) for a minimum of four (4) hours without evidence of external damage (see 6.4) and shall be tested as specified in 4.5.5.2 for compliance with 3.7.1 and 3.9.

*3.10.3 Vibration.

3.10.3.1 Transportation vibration at high temperature. The sequential timer shall withstand at 140 ± 5°F vibration tests along each of its three (3) mutually perpendicular axis. The vibration schedule shall be 10-2000-10 hertz at 0.015 inch double amplitude (da) or 15 input peak gravity units (g) which ever is less for three minutes (3) total, 1.5 minutes up and 1.5 minutes down, twice in each axis. The timer shall be tested as specified in 4.5.5.3.1.

3.10.3.2 Operational vibration. The sequential timer shall withstand vibration tests along each of its three (3) mutually perpendicular axis as specified below without evidence of external damage. The timer shall be tested as specified in 4.5.5.3.2.

a. 0.3 g2/hertz (Hz) power spectral density (19.0 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 12 decibels per octave so that the total frequency enveloped is 50 to 2000 Hz for 10 seconds. The tolerance level shall be ± 3 decibels (Db) between 100 to 1000 Hz and ± 6 Db between 1000 to 2000 Hz.

b. 0.06 g2/Hz power spectral density (8.5 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 12 Db do that the frequency enveloped is 50 to 2000 Hz for one (1) minute. The tolerance level shall be ± 3 Db between 100 to 2000 Hz.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

3.10.4 Shock.

3.10.4.1 Mechanical shock. The sequential timer shall withstand three (3) shocks of 50 ± 5 gravity units peak in each direction along its three (3) mutually perpendicular axis without evidence of external damage (see 6.4). Shock pulses shall be 11 ± 0.6 milliseconds (ms) in duration. The timer shall be tested as specified in 4.5.5.4.1.

*3.10.4.2 Thermal shock. The sequential timer shall withstand three alternate exposures of 140° ± 5°F and -40° ± 5°F without evidence of external damage (see 6.4) when tested as specified in 4.5.5.4.2.

3.10.5 Acceleration. The sequential timer shall withstand steady acceleration of 50 ± 5 g's for one (1 ± 0.1) minute in each direction along its three (3) mutually perpendicular axis without evidence of external damage (see 6.4). The timer shall be tested as specified in 4.5.5.5.

*3.11 Operating torque. When switches are tested in accordance with 4.6, the torque required to rotate each switch shaft between any detent position shall be 8 ± 2 pound-inches.

*3.12 Start Signal Sensitivity. The timer shall be capable of rejecting any start signals (open relay contacts) of less than 50 ± 5 milliseconds in duration. Testing shall be as specified in 4.7.

*3.13 Soldering. All soldering shall be in accordance with MIL-STD-2000.

*3.14 Unit of product. The unit of product as tabulated below shall have complied to their respective specification before they are assembled to the timer. Objective evidence (see 6.6) providing compliance with these specifications shall be made available to the Government.

Product Dwg No. or Type	Specification
Encoder Module 12551110	MIL-STD-883, Level B

3.15 Workmanship.

3.15.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, cold shuts, or any similar characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

3.15.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.15.3 Burrs. Parts shall be free from burrs.

3.15.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.15.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.15.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection and standards quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

*4.3.1 Quality to be submitted. When the contract requires first article inspection, the contract shall submit 8 complete assemblies (see dwg 12551040) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

*4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg and to the tests and inspections specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be from the same source as will be used in regular production.

TABLE III. - First article inspection and tests

Inspection or Test		Dest	Non-Dest
1.	Classification of characteristics (see 4.4.2.1 thru 4.4.2.16)		X
2.	Continuity loop (see 3.6)		X
3.	Input current (see 3.7.1)		X
4.	Output current (see 3.8)		X
5.	Signal output (see 3.9)		X
6.	Operating torque (see 3.11)		X
7.	Low temperature (see 3.10.1)		X
8.	High temperature (see 3.10.2)		X
9.	Thermal shock (see 3.10.4.2)	X	
10.	Transportation vibration (see 3.10.3.1)	X	
11.	Operational vibration (see 3.10.3.2)	X	
12.	Mechanical shock (see 3.10.4.1)	X	
13.	Acceleration (see 3.10.5)		X
14.	Start Signal Sensitivity (see 3.12)		X

During the first article test, additional Signal Output tests (see 3.9) shall be performed at the start of the low and high temperature soak periods and, also, after transportation vibration 3.10.3.1 when the unit has returned to room temperature.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

*4.3.3 Inspection equipment. The inspection and testing specified in 4.3.2 shall be accomplished by using the gages specified in 4.4.2.1 and acceptance test equipment, operation instruction, and calibration instructions approved according to provisions in MIL-A-48078.

4.3.4 Rejection. If any item fails to comply with any of the applicable requirement, the first article sample shall be rejected.

4.4 Quality conformance inspection.

*4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

*4.4.2 Examinations and tests. See MIL-A-48078 and the Quality Conformance Inspection Tables and 4.4.3 herein.

a. Critical defects. Critical defects shall be inspected one hundred percent.

b. Major defects. Unless otherwise indicated Major defects shall be sampled in accordance with Table IV.

c. Minor defects. Unless otherwise indicated Minor defects shall be sampled in accordance with Table V.

*4.4.2.1 Classification of defects. Critical, major, and minor defects are defined in MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

TABLE IV. - Major characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	212-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	162	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

TABLE IV. - Major characteristics (con't)

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

TABLE V. - Minor characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH * 4.4.2.1	TITLE: Processor Printed Wiring Board		SHEET 1 OF 1		DRAWING NUMBER 12551050. NEXT HIGHER ASSEMBLY 12551049
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Diameter, max, of mounting holes, incorrect		Table IV	3.1	Gage
102.	Location, max, of mounting holes, incorrect		Table IV	3.1	Gage
103.	Solder plating must be completely present on each conductor path and pad		Table IV	3.1	Visual
104.	Solder plating must be completely present on all through holes		Table IV	3.1	Visual*
105.	Blisters or wrinkles in conductor path and pad		Table IV	3.1	Visual*
106.	Separation of the plated-through-holes from the conductor path		Table IV	3.1	Visual*
107.	Separation of the conductor path from the base material		Table IV	3.1	Visual*
108.	Delamination of base material		Table IV	3.1	Visual*
109.	Voids or cracks in excess of 10% of thru hole surface area		Table IV	3.1	Visual
110.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:					
201.	Plating thickness (including thru holes), incorrect		Table V	3.1	Gage

NOTES:
*Inspect using a suitable 10 power magnification microscope.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Firing Printed Wiring Board		12551053
			NEXT HIGHER ASSEMBLY 12551052

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Diameter, max, of mounting holes, incorrect		Table IV	3.1	Gage
102.	Location, max, of mounting holes, incorrect		Table IV	3.1	Gage
103.	Solder plating must be completely present on each conductor path and pad		Table IV	3.1	Visual
104.	Solder plating must be completely present in all through holes		Table IV	3.1	Visual*
105.	Blisters or wrinkles in conductor path and pad		Table IV	3.1	Visual*
106.	Separation of the plated-through-holes from the conductor path		Table IV	3.1	Visual*
107.	Separation of the conductor path from the base material		Table IV	3.1	Visual*
108.	Delamination of base material		Table IV	3.1	Visual*
109.	Voids or cracks in excess of 10% of thru hole surface area		Table IV	3.1	Visual
110.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:					
201.	Plating thickness (including thru holes), incorrect		Table V	3.1	Gage

NOTES:
*Inspect using a suitable 10 power magnification microscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET		DRAWING NUMBER
4.4.2.3	Switch Printed Wiring Board		1 OF 1		12551061
					NEXT HIGHER ASSEMBLY
					12551060
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Diameter, max, of mounting holes, incorrect		Table IV	3.1	Gage
102.	Location, max, of mounting holes, incorrect		Table IV	3.1	Gage
103.	Solder plating must be completely present on each conductor path and pad		Table IV	3.1	Visual
104.	Solder plating must be completely present on all through holes		Table IV	3.1	Visual*
105.	Blisters or wrinkles in conductor path and pad		Table IV	3.1	Visual*
106.	Separation of the plated-through-holes from the conductor path		Table IV	3.1	Visual*
107.	Separation of the conductor path from the base material		Table IV	3.1	Visual*
108.	Delamination of base material		Table IV	3.1	Visual*
109.	Voids or cracks in excess of 10% of thru hole surface area		Table IV	3.1	Visual
110.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:					
201.	Plating thickness (including thru holes), incorrect		Table V	3.1	Gage

NOTES:

*Inspect using a suitable 10 power magnification microscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551082
4.4.2.4	Connector				NEXT HIGHER ASSEMBLY 12551084
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Mislocation of pins (2) from Master keyway		Table IV	3.1	Gage
102.	Length from mounting surface		Table IV	3.1	Gage
103.	Failure to fit with mating connector		Table IV	3.1	Gage
104.	Missing parts		Table IV	3.1	Visual
105.	Manufacturer's identification number missing, illegible, or incorrect		Table IV	3.1	Visual
106.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551049
4.4.2.5	Processor Board Assembly (After Assembly of IC)				NEXT HIGHER ASSEMBLY 12551084
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Designated ic oriented, incorrect		Table IV	3.2	Visual
102.	Ic positioned on board, incorrect		Table IV	3.2	Visual
103.	Lead extension beyond width of conductive path		Table IV	3.13	Visual
104.	Lead contact with conductive path incorrect		Table IV	3.13	Visual*
105.	Solder fillet length is less than twice width of path		Table IV	3.13	Visual*
106.	Ic damaged		Table IV	3.15	Visual*
107.	Board delaminated, measled (more than 50% of the distance between adjacent conductive paths), crazed, and/or blistered		Table IV	3.15	Visual*
108.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:

*Inspect using a suitable 10 power magnification microscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	*Firing Board Assembly (After Assembly of IC)				12551052
					NEXT HIGHER ASSEMBLY 12551084
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Designated ic oriented, incorrect		Table IV	3.2	Visual
102.	Ic positioned on board, incorrect		Table IV	3.2	Visual
103.	Lead extension beyond width of conductive path		Table IV	3.13	Visual
104.	Lead contact with conductive path incorrect		Table IV	3.13	Visual*
105.	Solder fillet length is less than twice width of path		Table IV	3.13	Visual*
106.	Ic damage		Table IV	3.15	Visual*
107.	Board delaminated, measled (more than 50% of the distance between adjacent conductive paths), crazed, and/or blistered		Table IV	3.15	Visual*
108.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:
*Inspect using a suitable 10 power magnification microscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551060
4.4.2.7	Switch Board Assembly				NEXT HIGHER ASSEMBLY 12551083
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Designated ic oriented, incorrect		Table IV	3.2	Visual
102.	Ic positioned on board, incorrect		Table IV	3.2	Visual
103.	Lead extension beyond width of conductive path		Table IV	3.13	Visual
104.	Lead contact with conductive path incorrect		Table IV	3.13	Visual*
105.	Solder fillet length is less than twice width of path		Table IV	3.13	Visual*
106.	Ic damage		Table IV	3.15	Visual*
107.	Board delaminated, measled (more than 50% of the distance between adjacent conductive paths), crazed, and/or blistered		Table IV	3.15	Visual*
108.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:

*Inspect using a suitable 10 power magnification microscope.

Source: https://assist.dla.mil -- Downloaded: 2022-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551081
4.4.2.8	Plate, Switch Mounting				NEXT HIGHER ASSEMBLY 12551083
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Location of Optical Encoder Switch mounting (12) and switch alignment holes (4)		Table IV	3.2	Gage
102.	Evidence of poor workmanship		Table IV	3.15	Visual
103.	Location and size of 7 face and edge fastening holes		Table IV	3.2	Gage
Minor:	None defined.				

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET		DRAWING NUMBER
4.4.2.9	Switch, Optical		1 OF 1		12551045
					NEXT HIGHER ASSEMBLY
					12551083
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Indexing torque on shaft, incorrect		Table IV	3.2	Gage
102.	Torque on 4 screws, incorrect		Table IV	3.2	Gage
103.	Torque on 3 screws, incorrect		Table IV	3.2	Gage
104.	Any components missing, damaged, or improperly assembled		Table IV	3.2	Visual
105.	Objective evidence of performance acceptance (dwg 12551110)		Table IV	6.6	Visual
106.	Evidence of poor workmanship		Table IV	3.16	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Knob		12551044
			NEXT HIGHER ASSEMBLY
			12551068 (see notes)

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Size and positioning of square and "D" holes		Table IV	3.2	Gage
102.	Thickness of skirt		Table IV	3.2	Gage
103.	Size and position of pin shaft		Table IV	3.2	Gage
104.	Location of notch		Table IV	3.2	Gage
105.	Annodize finish evenly applied		Table IV	3.1	Visual
Minor:	None defined.				

NOTES:
*Additional next higher assemblies: 12551072 and 12551073

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551080
4.4.2.11	Plate, Front				NEXT HIGHER ASSEMBLY 12551083
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Location, diam, max, of four switch knob holes		Table IV	3.2	Gage
102.	Location, diam, max, of four switch reading holes		Table IV	3.2	Gage
103.	Overall height, incorrect		Table IV	3.2	Gage
104.	Location, diam, max, of seven mounting holes		Table IV	3.2	Gage
105.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET		DRAWING NUMBER
4.4.2.12	Flex Cable Assembly		1 OF 1		12551064, 12551066 12551065
					NEXT HIGHER ASSEMBLY 12551084
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Hole pattern accuracy		Table IV	3.2	Gage
102.	Overall dimensions		Table IV	3.2	Gage
103.	Strain relief bar(s) positioning		Table IV	3.2	Gage
104.	Proper marking		Table IV	3.2	Visual
105.	Surface solderability		Table IV	3.13	Visual
106.	Conduction patterns		Table IV	3.2	Visual
107.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET		DRAWING NUMBER
4.4.2.13	Flex Cable Assembly		1 OF 1		12551067
					NEXT HIGHER ASSEMBLY
					12551084
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Hole pattern accuracy		Table IV	3.2	Gage
102.	Overall dimensions		Table IV	3.2	Gage
103.	Strain relief bar(s) positioning		Table IV	3.2	Gage
104.	Connector potting size proper		Table IV	3.2	Gage
105.	Surface solderability		Table IV	3.13	Visual
106.	Conductor patterns		Table IV	3.2	Visual
107.	Proper marking		Table IV	3.2	Visual
108.	Evidence of poor workmanship		Table IV	3.15	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Knob Assembly	1 OF 1	12551068, 12551073 12551072
			NEXT HIGHER ASSEMBLY 12551083

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Letters and numbers legible		Table IV	3.2	Visual
102.	Letters and numbers correctly aligned		Table IV	3.2	Gage
Minor:	None defined.				

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Housing	1 OF 1	12551042
			NEXT HIGHER ASSEMBLY: 12551040

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Diameter, max, or four mounting holes, incorrect		Table IV	3.2	Gage
102.	Location of four mounting holes, incorrect		Table IV	3.2	Gage
103.	Location of one receptacle alignment holes, incorrect		Table IV	3.2	Gage
104.	Diameter, max, of one receptacle alignment holes, incorrect		Table IV	3.2	Gage
105.	Depth, min, of one receptacle alignment holes, incorrect		Table IV	3.2	Gage
106.	Pin missing in two receptacle alignment holes		Table IV	3.2	Visual
107.	Thickness of base (Surface 'A'), incorrect		Table IV	3.2	Gage
108.	Objective evidence of radiographic inspection		Table IV	6.6	Visual
109.	Evidence of poor workmanship		Table IV	3.15	Visual
110.	Location of Optical Encoder Switch mounting (62) holes		Table IV	3.2	Gage
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Timer, Sequential	1 OF 1	12551040
			NEXT HIGHER ASSEMBLY: --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Component damaged or loose			3.2	Visual
102.	Component missing or improper			3.2	Visual
103.	Improper assembly of component			3.2	Visual
104.	Identification plate missing, illegible or improper			3.2	Visual
105.	Knob numerals and letter improper			3.2	Visual
106.	Failure to fit with mating connector			3.2	Gage
107.	Height (mtg surface to mtg surface) max, incorrect			3.2	Gage
108.	Torque on four (4) rear terminal board nuts (MS21043-06) incorrect			3.2	Gage
109.	Torque on receptacle nut incorrect			3.2	Gage
110.	Receptacle nut below housing surface			3.2	Visual
111.	Receptacle extension, incorrect			3.2	Gage
112.	Location mtg holes (8), incorrect			3.2	Gage
113.	Serialization missing or improper (see 3.4)			3.4	Visual
114.	Torque seven (7) front plate screws and six (6) housing assy screws (MS24693-C25)			3.2	Gage
115.	Evidence of poor workmanship			3.15	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-45886A(AR)

PARAGRAPH *	TITLE		SHEET		DRAWING NUMBER
4.4.2.17	Preparation for Delivery		1 OF 1		--
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101.	Preservation packing and packaging not in accordance with ASTM-D 3951-82		Table IV	5.2	Visual
102.	Marking not in accordance with ASTM-D 3951-82		Table IV	5.3	Visual
Minor:					
201.	Evidence of poor workmanship		Table V	3.15	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T04:05Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity loop (see 3.6)	Major
b. Input current (see 3.7.1)	Major
c. Output current (see 3.8)	Major
d. Operating torque (see 3.11)	Major
e. Thermal shock (see 3.10.4.2)	Major
f. Transportation vibration (see 3.10.3.1)	Major
g. Signal output (see 3.9)	Major

NOTE: It is permissible to incorporate output current test as part of the thermal shock test.

4.4.3.1 Continuity loop, input current, output current, thermal shock, transportation vibration, and signal output. Each sequential timer shall be tested in accordance with 3.7.1, 3.8, 3.10.4.2, 3.10.3.1 and 3.9, if any defect is found, the item shall be rejected and removed from the lot.

*4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see MIL-A-48078 and 6.5 herein).

4.5 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

*4.5.1 Continuity loop. The continuity loop resistance shall be measured with an approved milliohmeter or approved automated test equipment for compliance with 3.6.

4.5.2 Input power. The input power for the integrated circuit portion of each channel shall be measured as illustrated in Figure 1 for compliance with 3.7. The maximum resistance of each current measuring device shall be 0.1 ohm.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

*4.5.3 Output current. The sequential timer shall be placed into a temperature chamber. The temperature shall be raised to 140 ± 5°F and held at the specified temperature for 2 hours minimum, or until the internal temperature of the timer has stabilized (see 6.3). While maintained at this temperature, the timer shall be evaluated for compliance with 3.8, using test cycle "P" from Table II of 3.9. The above procedure is to be repeated at -40° ± 5°F. The output current shall be measured with an approved current measuring device or controlled with calibrated/approved acceptance test equipment for unit acceptance. When the test is performed at 140°F, the input voltage shall be 7 + 1/4 - 0 vdc, and when performed at -40°F, the input voltage shall be 6 plus 1/4 minus 0 vdc.

4.5.4 Signal output. the signal output time pulse for each setting illustrated in Table II shall be measured with an approved electrical measuring device for compliance with 3.9. The input voltage shall be 6.5 ± 1/4 vdc. The current of each output pulse shall be as shown in paragraph 3.8 Table I.

4.5.5 Environmental testing (see 3.10).

4.5.5.1 Low temperature. The sequential timer shall be placed into a temperature chamber. The temperature shall be lowered to -40 ± 5°F and held at the specific temperature for four (4) hours, minimum or until the timer has stabilized (see 6.3). While maintained at this temperature, the timer shall be evaluated for compliance with 3.7 and 3.9, except at an input voltage of 6 + 1/4 - 0 vdc. The first three (3) digits of the test number shall be changed to 070.

*4.5.5.2 High temperature. The sequential timer shall be placed into a temperature chamber. The temperature shall be raised to 140 ± 5°F and held at the specific temperature for four (4) hours, minimum or until the timer has stabilized (see 6.3). While maintained at this temperature, the timer shall be evaluated for compliance with 3.7 and 3.9, except at an input voltage of 7 + 1/4 - 0 vdc. The first three (3) digits of the test number shall be changed to 080.

4.5.5.3 Vibration.

*4.5.5.3.1 Transportation vibration at high temperature. The sequential timer shall be mounted securely to a fixture, capable of being readily positioned along each of its three (3) mutually perpendicular axis, and placed in a suitable vibrator-temperature chamber. The temperature shall be raised to 140 ± 5°F and held at this temperature for four (4) hours minimum, or until the internal timer temperature has stabilized (see 6.3). The timer shall then be subjected to the vibration level specified in 3.10.3.1. During the vibration test, the timer

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

shall be tested for compliance with 3.9 per test cycle "P", and the input voltage for compliance be 7 + 1/4 - 0 vdc. The first three (3) digits of the test number shall be changed to 090, 091, and 092 for the X, Y, and Z axis respectively. Refer to figure 4 for axis designation.

4.5.5.3.2 Operational vibration. The sequential timer shall be mounted securely onto a fixture capable of being readily positioned along each of its three (3) mutually perpendicular axes. The mounted timer shall then be mounted on a vibrator and vibrated to the levels specified in 3.10.3.2. During the vibration, the timer shall be tested for compliance with 3.9 per test cycle "P". The first three (3) digits of the test number shall be changed to 190, 191, and 192 for the X, Y, and Z axis respectively of test (b). Refer to Figure 4 for axis designation.

4.5.5.4 Shock.

4.5.5.4.1 Mechanical Shock. The sequential timer shall be mounted securely onto a fixture capable of being readily positioned along each direction along its three (3) mutually perpendicular axis. The mounted timer shall then be mounted on a shock producing device and tested to the levels specified in 3.10.4 for a total of 18 shocks. Upon completion of this test, the timer shall be evaluated for compliance with 3.7 and 3.9. The first three (3) digits of the test number shall be changed to 101 for the six (6) directions along the "X" axes, 111 for the six (6) directions along the "Y" axis, and 121 for the six (6) directions along the "Z" axis. Refer to Figure 4 for the direction and axes designations.

*4.5.5.4.2 Thermal shock. The sequential timer shall be placed in an approved temperature chamber stabilized at 140° ± 5°F (see 6.3) and the timer shall remain at this temperature for a minimum period of two (2) hours. At the conclusion of this period, and within five minutes, the timer shall be transferred to another approved chamber stabilized at -40° ± 5°F (see 6.3) and the timer shall remain at this temperature for a minimum period of two (2) hours. This constitutes one cycle. The number of complete cycles shall be three. The timer shall be tested for compliance with 3.9 immediately after the two hour high temperature period in the third cycle. The cycle shall be resumed and the timer shall be tested for compliance with 3.9 immediately after the low temperature period in the third cycle. During the high temperature evaluation, the input voltage shall be 7 plus 1/4 minus 0 vdc. During the low temperature evaluation, the input voltage shall be 6 plus 1/4 minus 0 vdc. The first three digits of the test number shall be changed to 140 and 150 for high and low temperature, respectively. Return the timer to room temperature and test for compliance with 3.9 and the first three digits of the test number shall be changed to 160.

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

4.5.5.5 Acceleration. The sequential timer shall be mounted securely onto a fixture capable of being readily positioned along each of its three (3) mutually perpendicular axis. The mounted timer shall then be mounted on a centrifuge and accelerated to the levels specified in 3.10.5. Upon completion of this test, the timer shall be evaluated for compliance with 3.7 and 3.9. The first three (3) digits of the test number shall be changed to 130, 131, and 132 for the X, Y, and Z axis respectively. Refer to Figure 4 for the direction and axis designations.

*4.6 Operating torque. When the fully assembled switch is mounted in the timer housing, the torque required to rotate each switch shaft from one position to the next shall be measured and recorded for all positions in the clockwise and counterclockwise directions. The torque shall be within the limits specified in 3.11.

*4.7 Start Signal Sensitivity. The timer shall be set to test cycle A conditions as specified in 3.9. Power shall be normally closed start signal contact shown in Figure 1 shall be opened for 40 + 4 milliseconds interrupt by observing that the TDE(A) and TDE(B) output signals have occurred 0.5 ± 0.05 seconds later than specified. With power still applied to the timer, the start signal contact shall be opened for 60 ± 5 milliseconds, reclosed, and then opened again at 0.5 ± 0.05 seconds. TDE(A) and TDE(B) signal outputs shall occur in accordance with test cycle A requirements of 3.8. See figure 5 for graphic representation of the start signal sensitivity waveforms.

*5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be as specified (see 6.2.1).

5.1.1. The sequential timer shall be preserved and packaged in accordance with instructions contained in ASTM-D 3951-82, paragraph 5.1.

5.2 Packing.

5.2.1 The sequential timer shall be packed in accordance with the applicable instructions contained in ASTM-D 3951-82, paragraph 5.1.5.

5.3 Marking. Marking shall be in accordance with ASTM-D 3951-82, paragraph 5.1.5.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The sequential timers covered by this specification are intended for use in assembly of M239 adaption kit subassembly.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.3).
c. Applicable level of packaging and packing (see 5.0).

*6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423 Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR7-104.9 (n) (2) are involved and DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contractor purchase order requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph No.	Data requirements	Applicable DID No.
(a) 4.4.4	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-1714

(Copies of DID's required by contractors in connection with specific acquisition functions shall be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperatures plus tolerance specified.

6.4 External damage. Characteristics such as obliterated markings, damaged finish exposing base metal, rust and cracks that are detectable by visual examination, shall be considered cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

6.5 Equipment design approval. Submit equipment designs, as required, to Commander, U.S Army Armament Munitions and Chemical Command, ATTN: AMSMC-QAN-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, data item, DI-R-1714 will apply.

6.6. Objective Evidence.

6.6.1 General. Contractors inspection records and documents verifying compliance with applicable dwg. and specification requirements.

6.6.2 Materials. A statement (certification), supported by test data, that all materials produced or purchased by the contractor meets all requirements when such material is controlled by Government or commercial specifications referenced in contractual documents.

6.7 Changes from previous issue. The margins of this standard are marked with asterisks to indicate where significant changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

6.8 Subject term (key word) listing.

Signal output	Soldering
Vibration	Coded serialization
Shock	Flex cable
Acceleration	Optical encoder switch
Start signal sensitivity	Printed wiring board

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARRADCOM.

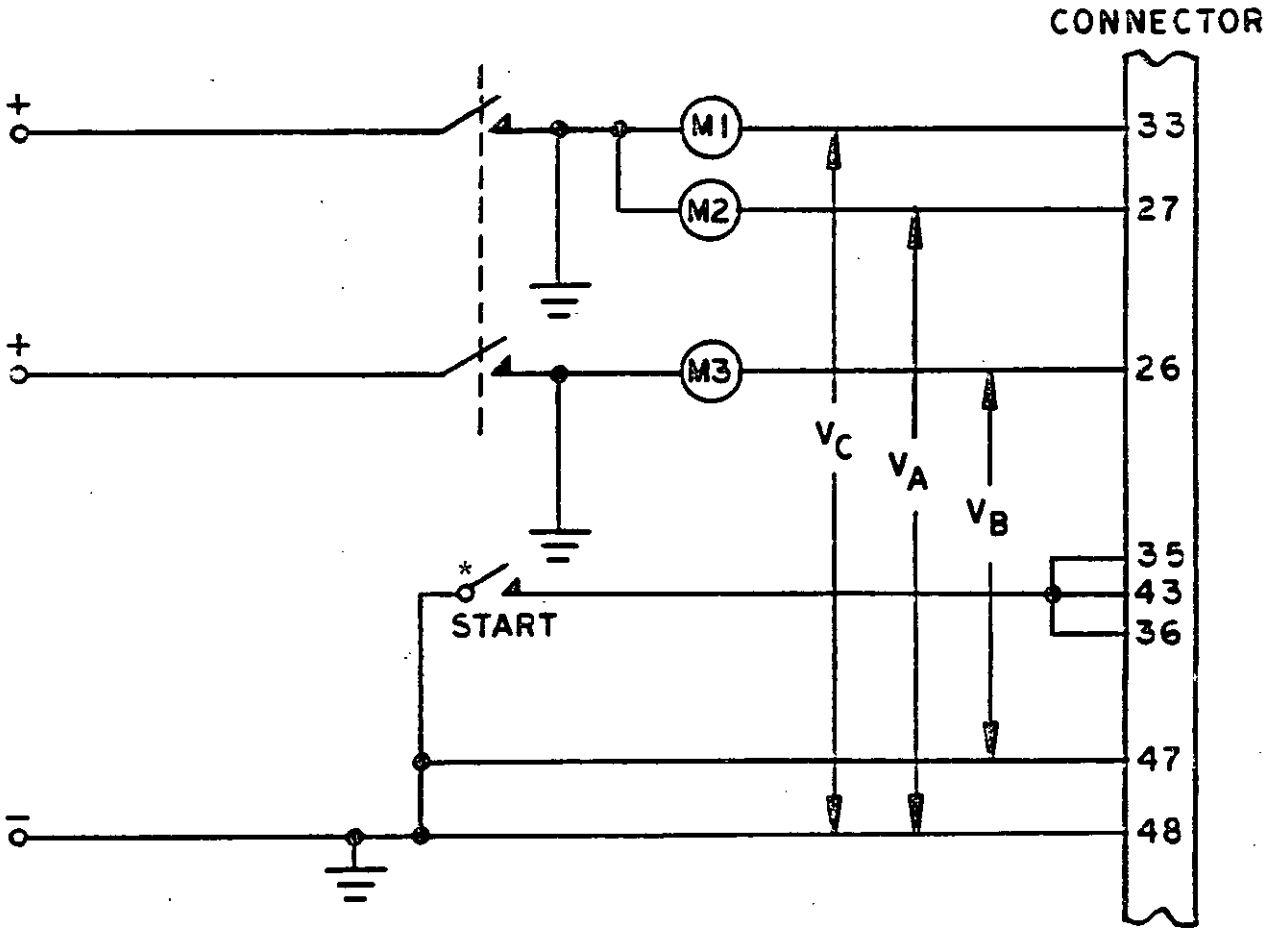
6.10 Supersession data. This specification supersedes the requirements of MIL-T-45886 (PA), dated 16 August 1976.

Custodian:
Army-AR

Preparing activity:
Army-AR

Project 1135-A304

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. THE TEST CABLE GROUND LEAD WIRES (9, 16, 30, 31, 47 AND 48) SHALL BE SHORTED AT THE TEST CONSOLE CONNECTOR.

2. *SIGNAL GROUND RELEASE FROM PINS 35, 36, 43 WITHIN 3 $\pm$ 2 SEC AFTER APPLICATION OF 6 VOLTS.

3. $V_A = 6.5\ ^{+0.75}_{-0.50}$ VDC, $V_B = 6.5\ ^{+0.75}_{-0.50}$ VDC, $V_C = 6.5\ ^{+0.75}_{-0.50}$ VDC

4. V_C SHALL BE MEASURED BETWEEN PINS 33 AND 48.

5. V_B SHALL BE MEASURED BETWEEN PINS 26 AND 47.

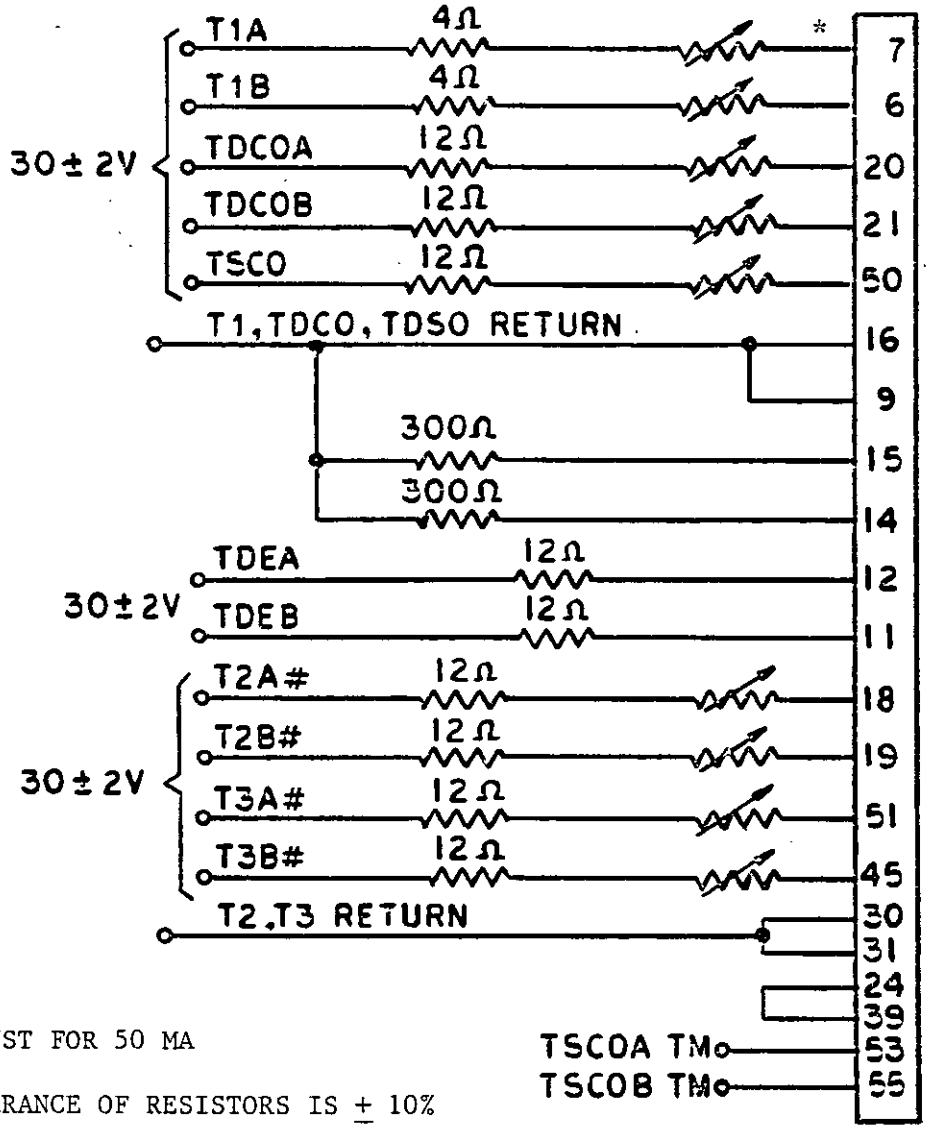
6. V_A SHALL BE MEASURED BETWEEN PINS 27 AND 48.

7. M1, M2, AND M3 ARE CURRENT MEASURING DEVICES AND MAY BE PART OF AUTOMATIC TEST EQUIPMENT.

CONNECTIONS FOR INPUT POWER

*FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.



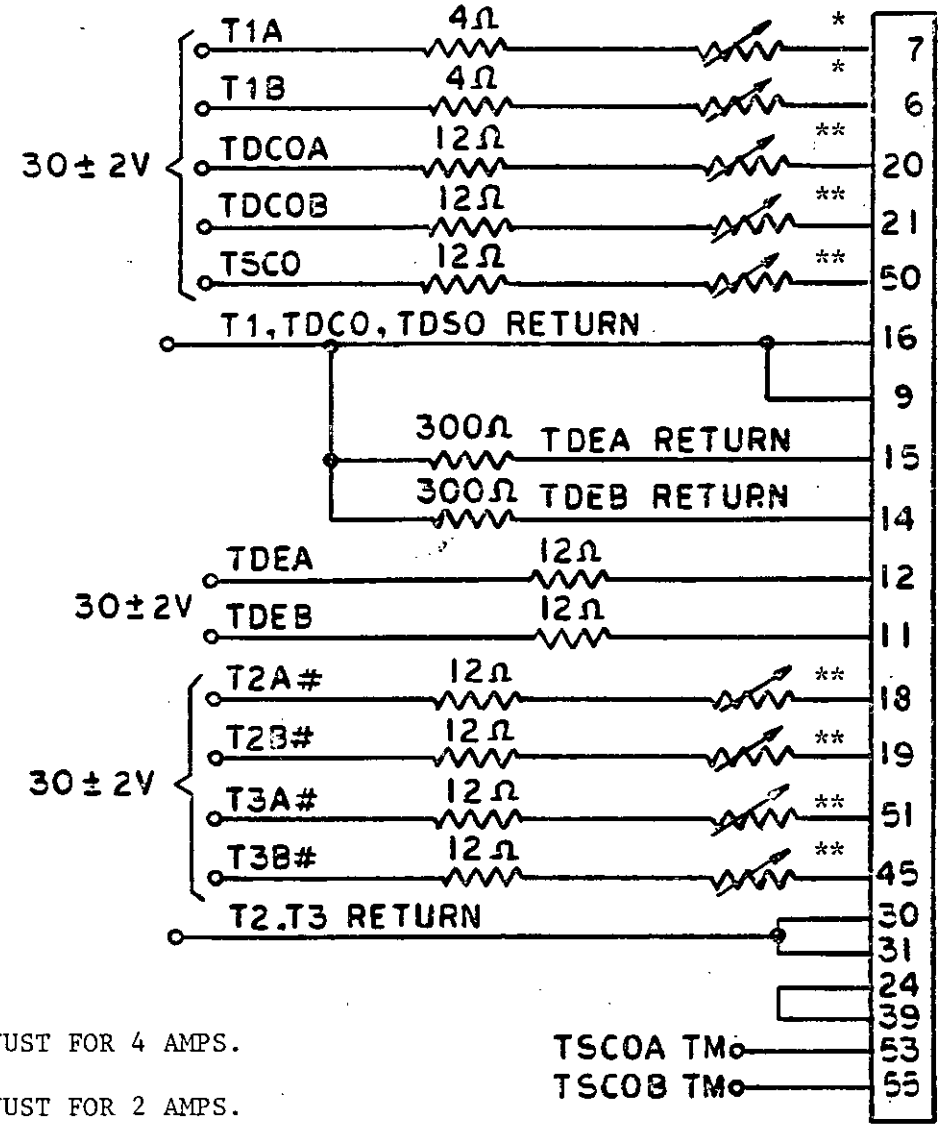
NOTES:

1. *ADJUST FOR 50 MA
2. TOLERANCE OF RESISTORS IS ± 10%
3. #30 VOLTS TO T2 AND T3 AFTER APPLICATION OF 6 VOLTS POWER AND BEFORE T1 OCCURS.
4. RESISTORS, 4 OHMS AND 12 OHMS, MAY BE INCLUDED IN LOAD RESISTANCE FOR TEST PURPOSES.

CONNECTIONS FOR SIGNAL OUTPUT

*FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. *ADJUST FOR 4 AMPS.

2. **ADJUST FOR 2 AMPS.

3. TOLERANCES OF RESISTORS IS ± 10%.

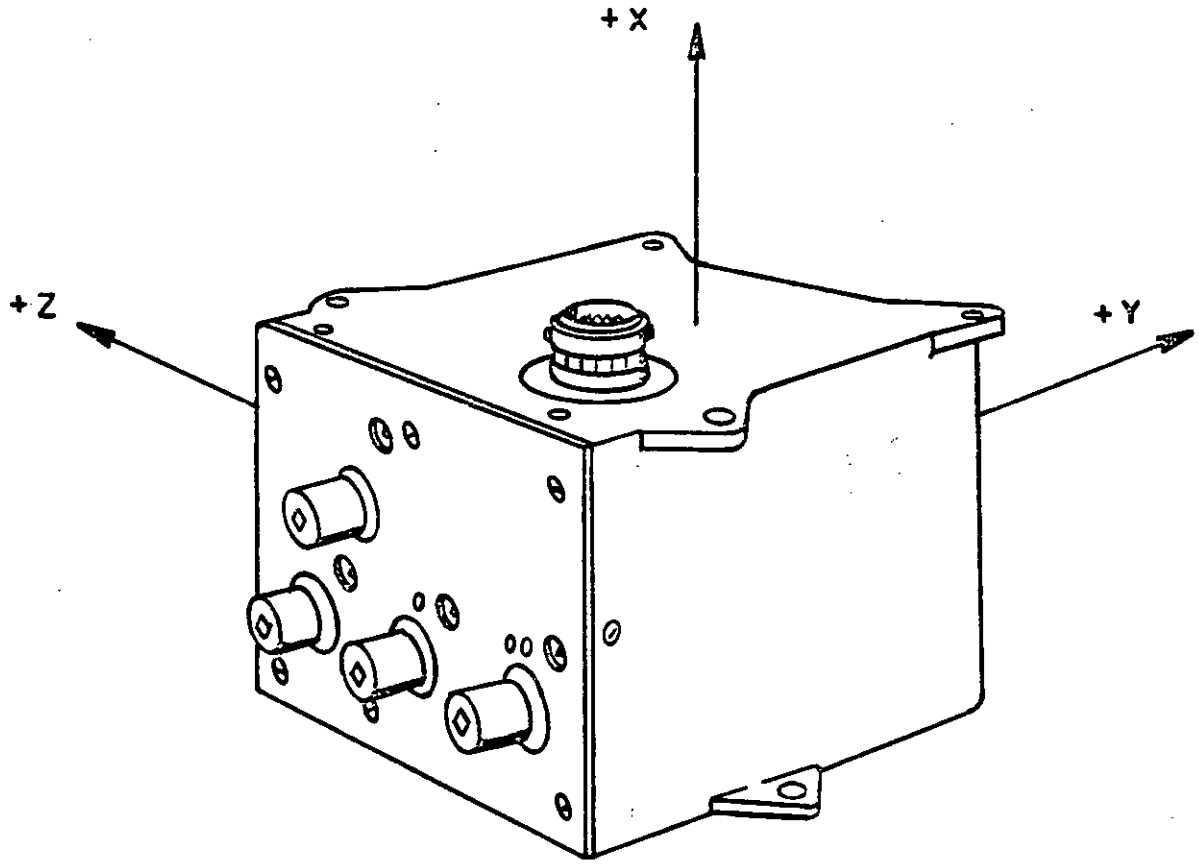
4. #30 VOLTS TO T2 AND T3 AFTER APPLICATION OF 6 VOLTS POWER AND BEFORE T1 OCCURS.

5. RESISTORS, 4 OHMS AND 12 OHMS, MAY BE INCLUDED IN LOAD RESISTANCES FOR TEST PURPOSES.

CONNECTIONS FOR OUTPUT CURRENT

*FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.

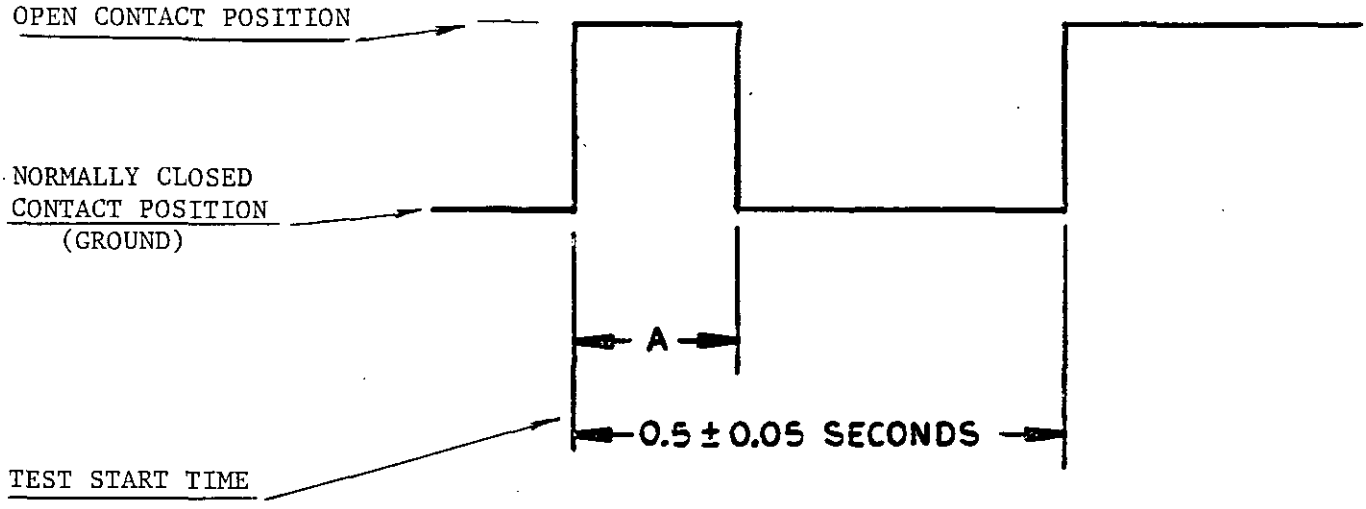


M42E2 SEQUENTIAL TIMER OUTLINE

AXES ORIENTATION

FIGURE 4

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the current version before use.



NOTES:

1. PULSE A = 40 ± 4 MILLISECONDS FOR CONFIRMATION OF "NO START" TESTING.
2. PULSE A = 60 ± 5 MILLISECONDS FOR CONFIRMATION OF PROPER "START" TESTING.

START SIGNAL SENSITIVITY TEST WAVE FORM

* FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Munitions &
Chemical Command
ATTN: AMSMS-QA
Picatinny Arsenal, NJ 07806-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-45886A (AR)	TIMER, SEQUENTIAL: M42E2

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(FOLD ON THIS LINE, OR ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T04:05Z
Check the source to verify that this is the currenversion before use.